Subfield parabolics and the corrected Suzuki classification

Human Compression project

Abstract

We complete the subgroup argument for configurations with a maximal subfield parabolic. A normalizer condition in a finite two-group forces every cover of a subfield point stabilizer into the central overgroup already treated by the correspondence with scalar subspaces. Thus that correspondence supplies the entire corrected family. We count its field orbits by an explicit formula for fixed projective lines under Frobenius. Together with the separately established normalizer families, this gives a corrected classification and a finite arithmetic formula for the total number of configurations. The numerical source tables and diagrams still require their separate complete audit; this draft does not by itself claim whole-source completion.

1 Acquired results and notation

Let $q = 2^n$, where $n \geq 3$ is odd, and let $G = \mathrm{Sz}(q)$. Write $a_\pm(q) = q \pm 2^{(n+1)/2} + 1$. We use the classical Suzuki partition, subgroup classification, ovoid action and automorphism group specified in the accepted note *Normalizer fibres in rank-two Suzuki geometries*. In particular every proper nonsolvable subgroup is a simple subfield group, all subfield groups of a prescribed order are conjugate, and their proper maximal simple subfield groups have prime relative field degree. The solvable maximal subgroups have already been treated, and a point stabilizer cannot be a parabolic in the geometries under consideration.

Fix a prime r dividing n, put $d = n/r \geq 3$, $s = 2^d$, $F = \mathbb{F}_q$, $K = \mathbb{F}_s$, and let $S = \mathrm{Sz}(s)$ be the standard maximal subfield subgroup. Its point stabilizer is $B_s = U_K H_K$, with $U_K = \{u(a,b) : a, b \in K\}$ and $H_K = \{h(k) : k \in K^*\}$. We retain the same faithful root-group coordinates as in the preceding notes:

$$u(a,b)u(c,e) = u(a+c, b+e+ac^\theta), \quad u(a,b)^{h(k)} = u(k^\theta a, k^{\theta+2}b), \qquad \theta = 2^{(n+1)/2}. \tag{1}$$

Put $P = U(F,F)$ and $Z_F = U(0,F)$. The two coordinate actions of K^* are the full scalar group of K. The automorphism $x \mapsto x^\theta$ on K has fixed field $\mathbb{F}_2$.

The accepted scalar-subspace note proves that the overgroups of B_s in $Z_F B_s$ are exactly

$$Q_T = Z_T B_s, \qquad K \leq T \leq F \text{ a } K\text{-subspace}, \quad Z_T = \{u(0,t) : t \in T\}. \tag{2}$$

A group Q_T covers B_s precisely when $\dim_K T = 2$; it then has order $s^3(s-1)$. The geometries $\Gamma(G; S, Q_T)$ are firm, connected, flag transitive, residually weakly primitive and locally doubly transitive. Their equivalence classes, in the source's convention of automorphisms of the specified group with type interchange allowed, are exactly the Frobenius orbits on the projective space $\mathbb{P}_K(F/K)$. These statements are reused, not proved twice. The task here is to prove exhaustion and compute those orbits.

2 Every point-stabilizer cover is central

Lemma 1. *Inside the point stabilizer $B_q = P \rtimes F^*$,*

$$\mathrm{N}_P(U_K) = Z_F U_K, \qquad \mathrm{N}_{B_q}(U_K) = Z_F U_K H_K. \tag{3}$$

Proof. An element $u(a,b) \in P$ normalizes U_K precisely when $ac^\theta + ca^\theta \in K$ for every $c \in K$, by (1). Taking $c = 1$ gives $a + a^\theta \in K$. Subtracting c times this relation from the general one gives $a(c^\theta + c) \in K$. Choose $c \in K \setminus \mathbb{F}_2$; its coefficient is nonzero, so $a \in K$. There is no restriction on b. This proves the first equality.

Conjugation by P does not change first coordinates. Therefore, if $u(a,b)h(t)$ normalizes U_K, its action on the first-coordinate image K gives $t^\theta K = K$, hence $t \in K^*$. Removing this diagonal factor reduces to the first equality. Conversely, all displayed elements normalize U_K. □

Theorem 2. *If $B_s < L < B_q$ and B_s is maximal in L, then $L = Q_T$ for a two-dimensional K-subspace T containing K. These are all proper subgroups outside S in which B_s is maximal.*

Proof. Put $V = L \cap P$, a normal two-subgroup of L containing U_K. If $V = U_K$, then $L \le \mathrm{N}_{B_q}(U_K)$. Equation (3) makes its image in B_q/P equal to H_K. Because $H_K \le L$, we obtain $L = VH_K = B_s$, a contradiction. Thus $V > U_K$.

The normalizer condition in a finite p-group gives $\mathrm{N}_V(U_K) > U_K$. For completeness, a proper subgroup H of a finite p-group acts on its cosets; the fixed cosets are $\mathrm{N}(H)/H$, all other orbits have length divisible by p, and the total index is divisible by p. There is a fixed coset, so there are at least p of them. This proves the condition. Now H_K normalizes both V and U_K, hence normalizes $\mathrm{N}_V(U_K)$. Therefore

$$B_s < \mathrm{N}_V(U_K)H_K \le L.$$

Maximality forces equality with L. Consequently U_K is normal in L, and (3) gives $L \le Z_F B_s$. The acquired correspondence (2) proves the claimed form and dimension.

Finally consider any proper overgroup L of B_s outside S. By the classical subgroup classification it either lies in a solvable normalizer or is a simple subfield group. Its large two-subgroup U_K excludes the cyclic torus normalizers. If it lies in a point stabilizer, a nonidentity involution of U_K forces that stabilizer to be the particular B_q. If $L = \mathrm{Sz}(t)$ and B_s is maximal in L, the classical maximal list forces B_s to be a point stabilizer of L, hence $t = s$. The accepted uniqueness theorem for a subfield overgroup of B_s would give $L = S$. Thus this case is excluded and the list already obtained is exhaustive. □

This strengthens the accepted correction: the groups Q_T are the entire replacement for the first row of source Table 8, not only a family contradicting that row. The printed group of order $qs(s-1)$ does not cover B_s when $r \ge 3$; the correct groups have order $s^3(s-1)$ and are parametrized as in (2).

3 The other subfield intersections

We shall use a further consequence of the normalizer incidence formula. If $\mathbb{F}_t \subseteq \mathbb{F}_s \subseteq \mathbb{F}_q$, every fixed $\mathrm{Sz}(t)$ is in exactly one $\mathrm{Sz}(s)$ of the prescribed order. Here and below both smaller field orders are at least eight. Indeed, all the subfield groups of each order are conjugate, both are self-normalizing by simplicity and the classical subgroup classification, and the subgroups of order $|\mathrm{Sz}(t)|$ inside $\mathrm{Sz}(s)$ form one conjugacy class. The incidence formula therefore gives one overgroup. This is a uniqueness assertion for each prescribed field order, not a claim that the entire interval has only one intermediate subgroup.

Theorem 3. *Besides the central covers in Theorem 2, all configurations with maximal parabolic $S = \mathrm{Sz}(s)$ are as follows:*

$J = S \cap L$	L	*condition*
$D_{2(s-1)}$	D_{2m}	$m/(s-1)$ *prime,* $m \mid q-1$
$C_{a_\delta(s)} \rtimes C_4$	$C_m \rtimes C_4$	$m/a_\delta(s)$ *prime,* $m \mid a_\epsilon(q)$
$\mathrm{Sz}(2^e)$	$\mathrm{Sz}(2^{ep})$	$n = erp$, r, p *distinct primes,* $e \geq 3$.

(4)

In the second row $\delta \in \{+1, -1\}$ and the ambient sign ϵ is the unique one for which $a_\delta(s) \mid a_\epsilon(q)$. Each permitted numerical choice in the first two rows gives one equivalence class. Each unordered pair of distinct prime indices in the last row gives one class.

Proof. The maximal-subgroup list in S leaves precisely the three intersection types in (4), after its point stabilizer. For $J = D_{2(s-1)}$, its odd rotation subgroup determines the unique split torus of G. An overgroup in a point stabilizer cannot contain the involution inverting that nontrivial odd subgroup: projection to the point stabilizer's abelian odd quotient would make inversion trivial. Coprimality excludes the nonsplit normalizers. Thus a solvable overgroup is in the unique split dihedral normalizer, and the cyclic maximal-subgroup calculation gives exactly the first row. A simple subfield overgroup in which J is maximal must have field order s; the already accepted uniqueness theorem would make it S. Within the ambient torus normalizer the group of each displayed order containing this fixed J is unique, so each choice gives one configuration class.

For $J = C_{a_\delta(s)} \rtimes C_4$, its characteristic odd subgroup similarly determines its unique ambient nonsplit torus. The point stabilizer and split normalizer are excluded by their odd orders. Its solvable covers are therefore exactly the prime torus extensions in the second row. A simple overgroup in which J is maximal must again have field order s: the two functions $a_+(2^d)$ and $a_-(2^d)$ strictly increase through nonoverlapping intervals as the odd exponent $d \geq 3$ increases, and the two signs at a fixed exponent are distinct. Hence its odd torus order determines s and δ. The uniqueness theorem then excludes it. For a fixed J each allowed m gives one group, proving the count one.

If $J = \mathrm{Sz}(2^e)$ is a simple subfield group, then $d/e = p$ is prime, by maximality in S. An overgroup L is nonsolvable, so is a subfield group $\mathrm{Sz}(2^{e\ell})$, with ℓ prime by maximality of J in L. Since $n/e = rp$, necessarily $\ell = r$ or p. If $\ell = p$, uniqueness gives $L = S$. Thus $\ell = r \neq p$. Conversely distinct r, p give two different maximal subfield groups over the fixed J. Their intersection is exactly J: it contains J, maximal in each, and neither contains the other. Their conjugacy class is unique by the same incidence argument, and type interchange identifies the two orders of the prime pair.

For the first two rows, the proposed L lies outside S and J is maximal in both; hence the intersection is exactly J. In every row the maximality of S gives generation of G and the rank-two coset criterion supplies firmness, connectivity, flag transitivity and residual weak primitivity. □

The nonoverlap used in the proof is elementary. For $s \geq 8$, $a_+(s) \leq 3s/2 + 1$, whereas $a_-(4s) = 4s - 2\sqrt{2s} + 1 > 3s/2 + 1$. Adjacent odd exponents multiply the field order by four, proving the assertion. The case $2^e = 2$ is not included in the simple-subfield row: $\mathrm{Sz}(2)$ is the solvable group $C_5 \rtimes C_4$, and any such maximal intersection is already governed by the nonsplit row.

4 Frobenius orbits of the central covers

The following argument is finite field algebra; it is independent of the subgroup classification. It applies to every odd $d \geq 1$ and odd prime r, with $n = dr$, even when r divides d.

Lemma 4. *For a cyclic finite-field extension E/D with generator τ, the elements of norm one are exactly $\tau(c)/c$ for $c \in E^*$. If the extension degree is odd in characteristic two, invariants are exact on every sequence of vector spaces with compatible actions: an invariant vector in a quotient has an invariant lift.*

Proof. The map $c \mapsto \tau(c)/c$ has kernel D^*, and its image has size $(|E|-1)/(|D|-1)$. The norm map on the cyclic group E^* is the power $(|E|-1)/(|D|-1)$, is onto D^*, and has a kernel of that same size. The image lies in the kernel by telescoping, proving equality. For the second assertion, average any lift over the acting cyclic group. Its odd order is invertible in characteristic two, and averaging preserves the specified quotient vector. □

Theorem 5. *For $0 \le j < n$, put $h = \gcd(d, j)$ and $g = \gcd(n, j)$, with the usual convention $\gcd(a, 0) = a$. The number b_j of K-projective lines in F/K fixed by ϕ^j is*

$$b_j = \begin{cases} \dfrac{2^{h(r-1)} - 1}{2^h - 1}, & g = rh, \\ \gcd(r, 2^h - 1) - 1, & g = h. \end{cases} \tag{5}$$

These are the only cases. Consequently the number of central cover classes is

$$C(n, d) = \frac{1}{n} \sum_{j=0}^{n-1} b_j. \tag{6}$$

Proof. Set $\sigma = \phi^j$ and $m = d/h$, the order of its restriction to K. Since $\gcd(d/h, j/h) = 1$,

$$g = h \gcd(r, j/h) \in \{h, rh\}.$$

This identity does not require r to be coprime to d. The order of σ on F is n/g.

Suppose $g = rh$. Then σ has order m on both fields. The invariant fields are $F^\sigma = \mathbb{F}_{2^{rh}}$ and $K^\sigma = \mathbb{F}_{2^h}$. Exactness by averaging gives

$$|(F/K)^\sigma| = 2^{h(r-1)}.$$

On a fixed K-line choose $v \neq 0$ with $\sigma(v) = \lambda v$. Applying $\sigma^m = 1$ gives $N_{K/\mathbb{F}_{2^h}}(\lambda) = 1$. Lemma 4 permits a scalar change making v fixed. Each fixed line then has exactly $2^h - 1$ nonzero fixed vectors. Dividing gives the first expression in (5).

Suppose $g = h$. The automorphism $\rho = \sigma^m = \phi^{jd/h}$ has order r on F and fixes K. Its one-eigenspace in F is exactly K. Since r is odd, trace identifies F/K with the trace-zero complement $V = \ker(\mathrm{Tr}_{F/K})$; this complement is also σ-invariant. The one-eigenspace is removed, so $V^\rho = 0$. On a σ-fixed K-line, σ acts semilinearly and ρ acts as a scalar $\mu = N_{K/\mathbb{F}_{2^h}}(\lambda) \in \mathbb{F}_{2^h}^*$. Necessarily $\mu^r = 1$ and $\mu \neq 1$.

Conversely, for every such μ, Lemma 4 applied to F/K gives a nonzero solution of $\rho(v) = \mu v$. Its solutions form one K-line: the ratio of two nonzero solutions is fixed by ρ. For $\mu \neq 1$ this line has trace zero, so lies in V. Commutation of σ and ρ sends the μ-eigenline to the $\sigma(\mu)$-eigenline. Thus the line is σ-stable exactly when $\mu \in \mathbb{F}_{2^h}$. There are $\gcd(r, 2^h - 1) - 1$ such nontrivial roots. This proves the second expression. Finally, counting the pairs consisting of an orbit and an automorphism fixing one of its points gives Burnside's formula (6). □

For example,

$$C(9, 3) = 1, \quad C(15, 3) = 41, \quad C(15, 5) = 3, \quad C(21, 3) = 1791, \quad C(21, 7) = 7.$$

The first two agree with the independently accepted literal projective-orbit certificates in the scalar-subspace note. The formula also covers the repeated prime index cases, such as $C(27, 9) = 19$; no coprimality of the absolute and relative field degrees has been imposed.

5 A corrected global list and total

Let $\omega(a)$ denote the number of distinct prime divisors of the positive integer a, with $\omega(1) = 0$. Each simple subfield case above has maximal absolute field exponent $d = n/r \geq 3$, for a prime divisor r of n. For each sign δ, let $t_\delta(q, s)$ be the unique ambient nonsplit torus order divisible by $a_\delta(s)$.

Theorem 6. *The complete classification consists of the solvable maximal parabolic families in the accepted normalizer note, together with (2) and (4), identifying a configuration when both descriptions give the same unordered pair. If n is composite, its total is*

$$T(n) = 1 + E(n) + \sum_{\substack{r \mid n \ \text{prime} \\ d = n/r \geq 3}} [C(n,d) + A(n,d)], \tag{7}$$

where

$$A(n,d) = \omega\left(\frac{q-1}{2^d - 1}\right) + \sum_{\delta = \pm 1} \omega\left(\frac{t_\delta(q, 2^d)}{a_\delta(2^d)}\right),$$
$$E(n) = \#\{\{r,p\} : r \neq p \text{ prime},\ rp \mid n,\ n/(rp) \geq 3\}.$$

If n is prime, put

$$v = \frac{q-2}{2n}, \qquad I = 1 + \frac{(q-2)(q+4)}{4n}, \qquad H = 1 + \frac{(q-2)(q+2)}{4n}.$$

Let η_0, η_+, η_- be the indicators that respectively $q-1, a_+(q), a_-(q)$ are prime. Then

$$T(n) = 1 + \eta_0\big[I + 2v + 1 + H(\omega(a_+(q)) + \omega(a_-(q)))\big] + \sum_{\epsilon = \pm 1} \eta_\epsilon\big[2v + (v+1)\omega(a_{-\epsilon}(q))\big] - \eta_+\eta_-(v+1). \tag{8}$$

Proof. Residual weak primitivity gives at least one maximal parabolic in G. The accepted preliminary obstruction excludes a point stabilizer. The accepted normalizer classification treats the three torus normalizers; Theorems 2 and 3 treat every maximal intersection in a subfield parabolic. These exhaust the classical maximal-subgroup list, proving the classification.

If n is composite, all three ambient odd factors are composite. The universal pair $(D_{2(q-1)}, q{:}(q-1))$ contributes one. All other families have a subfield parabolic. For a fixed s, the first two rows of (4) are counted by the distinct prime divisors of the indicated quotients, and the central family by $C(n,d)$. The last row has two maximal subfield parabolics and must be counted once for each unordered pair of distinct prime indices. Their simple intersection must have field order at least eight. Distinct field orders or distinct torus orders do not identify these families. Including the full ambient torus normalizer among the possible L in (4) already includes the corresponding family from the normalizer note, so it is not added again. This proves (7).

If n is prime there are no proper simple subfield groups. The split prime case adds the same-type contribution I, the four-group and cyclic-four contributions v and $v+1$, and one H for each prime divisor of either nonsplit order. For each prime nonsplit order there are two contributions v and one $v+1$ per prime divisor of the opposite order. When both nonsplit orders are prime, their common mixed family has been counted twice; subtract $v+1$. No other families overlap. This is (8). □

6 Local double transitivity and source disposition

The universal pair is locally doubly transitive: its local degrees are two and q, and the latter action is the affine action of $\mathbb{F}_q \rtimes \mathbb{F}_q^*$. Every corrected central cover is also locally doubly transitive by the acquired scalar-subspace theorem. Thus the source's claim that only the universal family survives this condition for $n > 5$ requires correction when n is composite.

There are no further locally doubly transitive families except, at $q = 8$, the pairs $(C_5 : C_4, C_5 : C_4)$ and $(C_5 : C_4, C_4 \times C_2)$. Here a cyclic complement of order four acts regularly on the four nonzero elements of C_5, and the other local action in the second pair has degree two. For the prime split families, an intersection of order two cannot be transitive on the $q-2 \geq 6$ other cosets. For a prime nonsplit normalizer with intersection C_4, double transitivity requires its prime torus order to satisfy $p-1 \leq 4$, so $p = 5$; the opposite torus cannot also have order five. For the remaining subfield families the local action of S already fails double transitivity. If J is a torus normalizer of S, then $|J| \leq 8s$ and $|S| > |J|(|J| + 1)$ for $s \geq 8$, so $[S : J] - 1 > |J|$. If $J = \mathrm{Sz}(t)$ is a proper simple subfield group, then $s \geq t^3$, $|J| < t^5$ and $|S| > s^5/2 > t^{10} + t^5$; the same inequality applies. A point stabilizer of size $|J|$ cannot act transitively on more than $|J|$ points.

This supplies a corrected form of source Lemma 6.9 and its Table 8, the complete family statement of Theorem 7.1, and the corresponding local double-transitivity column in Tables 9–11. The source's numerical Table 12 and its 29 small-group diagrams need explicit numerical and invariant joins. They are not silently certified by a general formula. The original source is D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie 39 (1998), no. 1, 97–120. Its retained structural inputs remain attributed to Suzuki, Tits and Lüneburg. The earlier component's phrase "right coset gJ" should read simply "coset gJ"; its equations are unaffected. No Lean or whole-original completion claim is made by this draft.